



Uganda National Roads Authority

Asset Management System

Bridge Management System (BMS) Data Dictionary

Asset Management System Documentation Map



This document contains the BMS Data Dictionary for the Uganda National Roads Authority (UNRA). It forms part of the Asset Management System documentation as shown on the previous page.

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SECTION 1

1. DEFINITION OF INVENTORY DATA FIELDS

1.1 BRIDGE

DESCRIPTION: This is a unique number assigned to each bridge.

FORMAT : Character String : Maximum length of 6 characters.

This is a compulsory field. It is very important to use the correct bridge number. This bridge number is the identification field in the Bridge Management System (BMS). Prefix "TB" will be bridges which do not have an official registered bridge number (#) yet. Prefix "X" has been reserved for Irish Crossings (drifts).

Examples:

B	0	0	1		
B	2	0	0		
X	1	0	0		
X	1	3	6		



1.2 BRIDGE NAME

DESCRIPTION: This item gives the name of the bridge.

FORMAT : Character String : Maximum length of 30 characters.

This is a compulsory field. In order to present the captured data more friendly this field must be filled in.

Examples: Lagwel
 Lakaye Bridge
 Kitgummatidi
 Nakironget Irish Crossing.

Note: If bridge does not have an official name, a name must be compiled by describing the features over and under the bridge.

1.3 TYPE CROSSING

DESCRIPTION: This field describes the intersecting features.

FORMAT : Integer String : The crossing type is described by a two digit code.

01 Road over river	08 Road over pedestrian
02 Road over rail	09 Agricultural underpass
03 Road over road	10 Road over Water pipes
04 Road over canal	11 Road over conveyer belt
05 Rail over road	12 Conveyer belt over road
06 Canal/Pipe over road	13 Other
07 Pedestrian over road	? Unknown

1.4 DESCRIPTION OF PRINCIPAL FEATURE

DESCRIPTION: This field describes the feature carried by the bridge.

FORMAT : Character String : Maximum length of 20 characters.

This is a compulsory field. If the feature carried by the bridge does not have an official name, the Link Name, or a description of the feature must be filled in.

Examples: Chosan-Amudat Road
Ariamoi Road
Moroto-Soroti Road
Nyakaita - Ibanda
Kahenge-Rwabigangura.

1.5 LINK NAME OF PRINCIPAL FEATURE

DESCRIPTION: This field indicates the link name of the road carried by the bridge.

FORMAT : Character String : Maximum length of 255 characters.

Example: Abim – Kilak, Aduku – Lira, etc.

Note: 1) Leave this field blank if the feature carried by the bridge is not a road.

1.6 KM DISTANCE ON PRINCIPAL FEATURE

DESCRIPTION: This field indicates the km distance of the link carried by the bridge.

FORMAT : Decimal String : Range between 0,00 and 999,99.

Km distances must always refer to the permanent RMS km reading. This could differ from the km distances used during the construction phase. For this reason, the km distance should be verified during inspections.

*Note: If the principal feature is not a road, this item must be left blank.
If the km distance is unknown, it must be indicated with a question mark.*

1.7 RIVER NAME

DESCRIPTION: This field indicates the name of the river flowing under the bridge.

FORMAT : Character String : Maximum length of 20 characters.

Example: River Kibimba; Malaba River; Katinda River; Mpanga River

*Note: If the river name is unknown, it must be indicated with a question mark.
If the secondary feature is not a river/stream, this field should be left blank.*

1.8 DESCRIPTION OF SECONDARY FEATURE

DESCRIPTION: This field describes of the feature passing under the bridge.

FORMAT : Character String : Maximum length of 20 characters.

This is a compulsory field. If the feature passing under the bridge does not have an official name, the Road Number, or a description of the feature must be filled in.

Examples: Rail
Pedestrian subway
Agricultural underpass .

1.9 LINK NAME OF SECONDARY FEATURE

DESCRIPTION: This field indicates the link name of the road passing under the bridge.

FORMAT : Character String : Maximum length of 255 characters.

Note: 1) If the feature passing under the bridge is not a road, this item must be left blank.

1.10 KM DISTANCE ON SECONDARY FEATURE

DESCRIPTION: This field indicates the km distance of the feature passing under the bridge.

FORMAT : Decimal Field : Range between 0,00 and 999,99.

Km distances must always refer to the permanent RMS km reading. This could differ from the km distances used during the construction phase. For this reason, the km distance should be verified during inspections.

Note: If the secondary feature is not a road, this item must be left blank.

1.11 RESPONSIBLE AUTHORITY (OWNER)

DESCRIPTION: This field indicates the name of the authority or institution that is responsible for the bridge.

FORMAT : Integer String : The owner is described by a two digit code.

01 UNRA
02
03
04
05
06 Municipal / Local Government
07 Private
08 Other
? Unknown

1.12 MAINTENANCE STATION

DESCRIPTION: This field gives the District in which the bridge is located.

The Maintenance Station is indicated by a two digit code:

01 Kampala	13 Gulu
02 Mpigi	14 Kitgum
03 Msaka	15 Moyo
04 Mubende	16 Arua
05 Luwero	17 Jinja
06 Mbarara	18 Tororo
07 Kabale	19 Mbale
08 Kasese	20 Soroti
09 Fort Portal	21 Moroto
10 Hoima	22 Kotido

11 Musindi	? Unknown
12 Lira	

1.13 1:50 000 TOPOGRAPHICAL MAP NUMBER AND NAME

DESCRIPTION: These fields give the 1:50 000 topographical map number and name.

FORMAT : Character Strings : Number : Maximum length of 6 characters.
Name : Maximum length of 20 characters.

1.14 GIS CO-ORDINATES (DECIMAL DEGREES)

DESCRIPTION: This field indicates the respective E and S co-ordinates of the centre point of the bridge in decimal degrees.

FORMAT : Decimal Field : Range between 0.00000 and 99.99999.

1.15 LAY-OUT PLAN

THE FOLLOWING DEFINITION IS VERY IMPORTANT!

Definition for Abutment (left) and Abutment (right)

When looking along the principal feature in the direction of increasing km distance, abutment (Left) will be behind you at the lower km distance and abutment (Right) will be in front of you at the higher km distance. If only the secondary feature has km distance markings, then abutment (Left) is to the left when viewing in the direction of increasing km distance and abutment (Right) to the right.

1.16 TYPE BRIDGE

DESCRIPTION: This field indicates the type of bridge according to a list of bridge types. (See Figure 1 on Page 7)

FORMAT : Integer String : The type bridge is described by a two digit code.

This is a compulsory field. Use the main span to define the bridge type of the bridge when the bridge consists of more than one type of span.

- | | |
|--|---|
| 01 Simply supported span | 06 Suspended span |
| 02 Continuous span | 07 Portal frame |
| 03 Double cantilevered span | 08 Cantilevered construction |
| 04 Single cantilevered span | 09 Strut frame |
| 05 Single suspended simply supported span | 10 Supported cantilevered construction |

11	Suspension bridge	22	Girdled frame
12	Three hinged arch rib	23	Cable stayed bridge
13	Three hinged arch slab	24	Box culvert (in-situ concrete)
14	Two hinged arch rib	25	Box culvert (precast concrete)
15	Two hinged arch slab	26	Concrete pipe culvert
16	Bowstring	27	Corrugated metal pipe culvert
17	Fixed arch slab	28	Corrugated metal arch (Armco) culvert
18	Fixed arch - deck at crown	29	Concrete arch culvert
19	Fixed arch - deck at intermediate level	98	Other
20	Lattice girder truss	?	Unknown
21	Vierendeel truss		

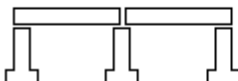
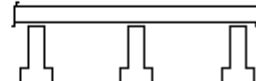
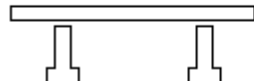
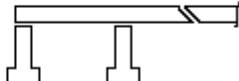
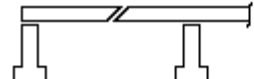
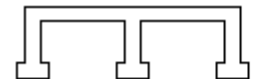
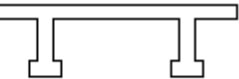
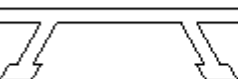
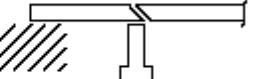
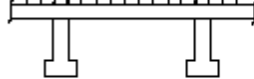
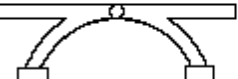
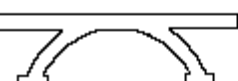
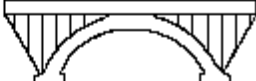
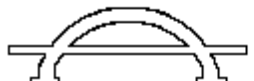
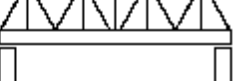
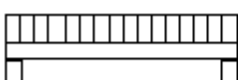
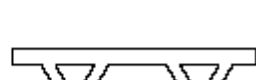
1 Simply supported span 	2 Continuous span 	3 Double cantilevered span 	4 Single cantilevered span 
5 Single suspended simply supported span 	6 Suspended span 	7 Portal frame 	8 Contilevered construction 
9 Strut frame 	10 Supported cantilevered construction 	11 Suspension bridge 	12 Threee hinged arch rib 
13 Three hinged arch slab 	14 Two hinged arch rib 	15 Two hinged arch slab 	16 Bowstring 
17 Fixed arch slab 	18 Fixed arch deck at crown 	19 Fixed arch deck at intermediate level 	20 Lattice girder truss 
21 Vierendeel truss 	22 Girdled frame 	23 Cable stayed bridge 	24 to 29 Culverts 

Figure 1: Type Bridge

1.17 TYPE DECK

DESCRIPTION: This field describes the type of deck configuration in a span. (See Figure 2 on Page 9)

FORMAT : Integer String : The Type Deck is described by a two digit code.

Use the main span's deck type when the bridge consists of more than one type of deck.

01 Solid slab	13 Multiple box girder Box beam
02 Voided slab	14 Twin box and slab
03 Inverted T-beams with infill	15 Multiple box and slab
04 Inverted T-beams (Pseudo box)	16 Steel I-beams encased in concrete
05 Box beams	17 Steel I-beam ribs (Jacked arch)
06 Beam and slab deck	18 Composite steel and concrete
07 Monolithic beam and slab deck	19 Non-composite steel and concrete
08 Rib deck	20 Steel with any other material
09 Voided spine beam	21 Solid slab with balustrade beam
10 Solid spine beam	22 Cell structure
11 Twin beam and slab	98 Other
12 Box girder – Box beam	? Unknown

1.18 TYPE DECK MATERIAL

DESCRIPTION: This field gives the type of material from which the deck is constructed.

FORMAT : Integer String : The type deck material is described by a two digit code.

Use the main span's type deck material when the bridge consists of more than one type of deck.

The type deck material is to be indicated with code “03” for in situ cast concrete cell structures and with “02” for precast units (cell structures).

01 Pre/post-stressed concrete	06 Steel and concrete
02 Precast units (cell structures)	07 Timber
03 Reinforced concrete	98 Other
04 Precast beams	? Unknown
05 Structural steel	

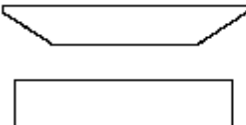
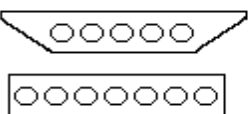
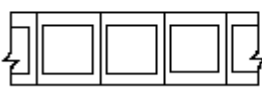
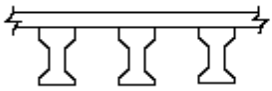
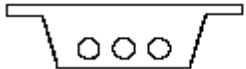
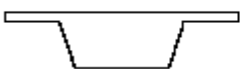
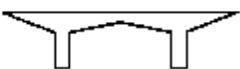
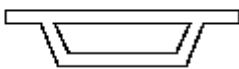
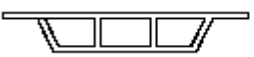
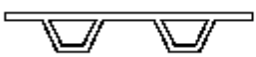
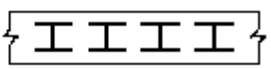
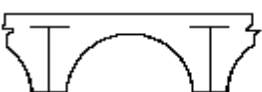
1 Solid slab 	2 Voided slab 	3 Inverted T-beams with infill 	4 Inverted T-beams (Pseudo box) 
5 Box beam 	6 Beam and slab 	7 Monolithic beam and slab 	8 Rib 
9 Voided spine beam 	10 Solid spine beam 	11 Twin beam and slab 	12 Box girder-Box beam 
13 Multiple box girder Box beam 	14 Twin box and slab 	15 Multiple box and slab 	16 Steel I-beams encased in concrete 
17 Steel I-beams ribs (Jacked arch) 	18 Composite steel and concrete 	19 Non-composite steel and concrete 	20 Steel with any other material 
21 Solid slab with balustrade beam 			

Figure 2: Type deck (Transverse sections)

1.19 TYPE ABUTMENT (LEFT/RIGHT)

DESCRIPTION: This field indicates the type of abutment. (See Figure 3 on Page 12)

FORMAT : Integer String : The type of abutment is described by a two digit code.

THE FOLLOWING DEFINITION IS VERY IMPORTANT!

Definition for Abutment (left) and Abutment (right)

When looking along the principal feature in the direction of increasing km distance, abutment (Left) will be at the lower km distance and abutment (Right) will be at the higher km distance. If only the secondary feature has km distance markings, then abutment (Left) is to the left when viewing in the direction of increasing km distance and abutment (Right) to the right.

- | | |
|---|--|
| 01 Mass concrete gravity type | 11 Counterfort RC |
| 02 Reinforced Concrete | 12 Cell structure |
| 03 R C wall with RC wingwalls | 13 Multiple V-type |
| 04 Spill through | 14 Perched abutment |
| 05 Seating beam/stub column | 22 Solid RC cantilever/wingwall |
| 06 Frame (only bridges, not cell structures) | 23 Solid RC cantilever/returnwall |
| 07 Buttressed concrete wall | 97 None |
| 08 Masonry wall | 98 Other |
| 09 Integral pile cap | ? Unknown |
| 10 Reinforced earth | |

1.20 TYPE PIERS

DESCRIPTION: This field gives the type of pier. (See Figures 4(a) on Page 13 and 4(b) on Page 14)

FORMAT : Integer String : The type pier is described by a two digit code.

This is a compulsory field. Use the pier type of the piers supporting the main span when there is more than one pier type.

- | | |
|--|-------------------------------------|
| 20 Solid R C wall | 28 Steel lattice |
| 21 Cellular R C Wall | 29 Single RC splayed pier |
| 22 Single R C column | 30 Multiple RC splayed piers |
| 23 Single RC hollow column | 31 Single RCV shape |
| 24 Multiple R C columns | 32 Multiple RCV shape |
| 25 Multiple R C columns (Beam on top) | 97 None |
| 26 Masonry wall | 98 Other |
| 27 Mass concrete wall | ? Unknown |

Note: RC = Reinforced Concrete

1.21 TYPE PARAPET/RAILING

DESCRIPTION: This field indicates the type of traffic barrier on the bridge. (See Figure 5 on Page 14)

FORMAT : Integer String : The type of traffic barrier or railing is described by a two digit code.

- | | |
|--|--|
| 01 Guard blocks / Guide blocks | 14 RC wall less than 300mm high plus balustrade |
| 02 Steel stanchion and railing (1) | 15 RC wall more than 300mm high plus balustrade |
| 03 Steel stanchion and railing (2) | 16 RC wall less than 300mm high plus guardrail |
| 04 RC stanchion / steel railing | 17 RC wall more than 300mm high plus guardrail |
| 05 RC stanchion / aluminium railing | 97 None |
| 06 RC wall less than 300mm | 98 Other |
| 07 RC 'New Jersey' balustrade | ? Unknown |
| 08 RC 'New Jersey' plus steel rail | |
| 09 RC solid balustrade | |
| 10 RC 'New Jersey' plus guardrail | |
| 11 Guardrail only | |
| 12 Steel stanchion and grid | |
| 13 RC wall more than 300mm high | |

Note: RC = Reinforced Concrete

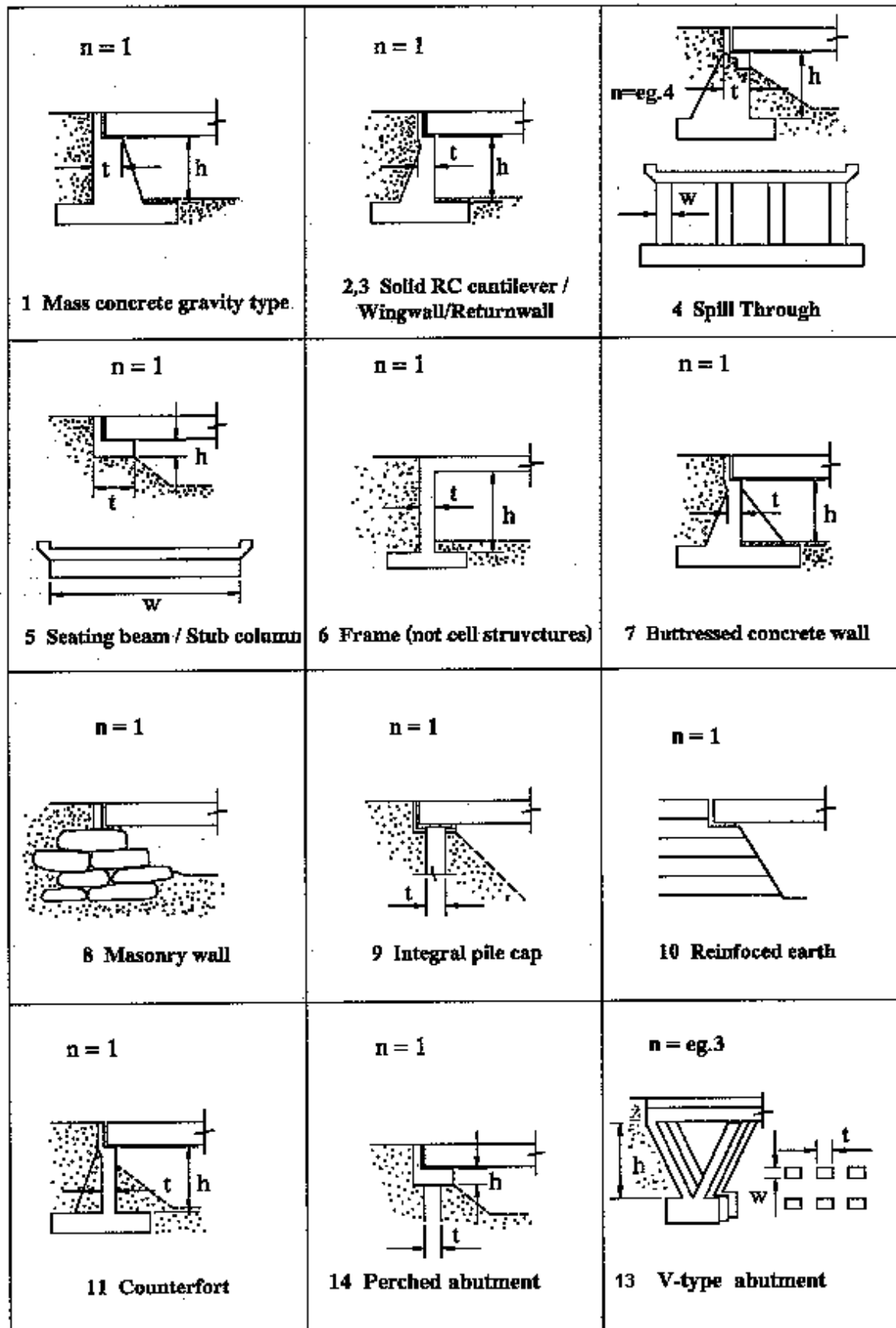


Figure 3 Type Abutment

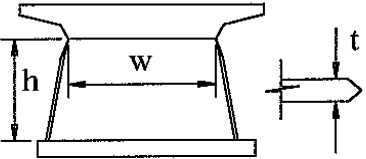
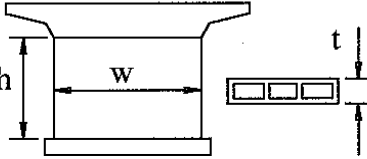
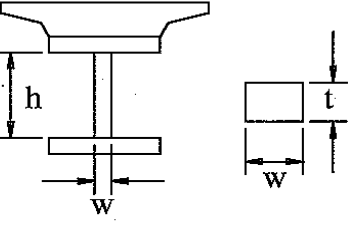
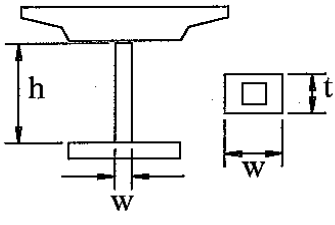
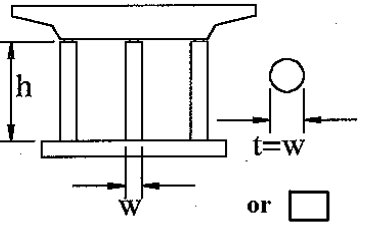
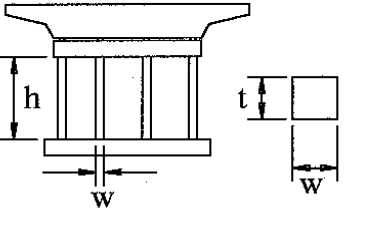
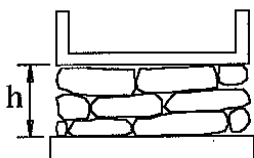
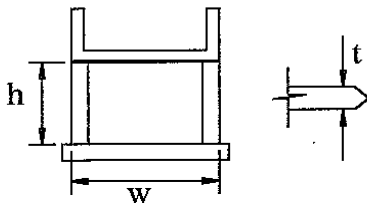
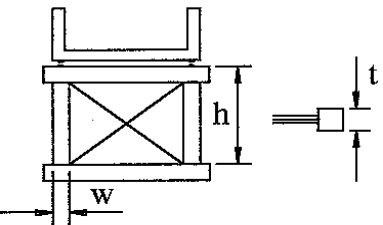
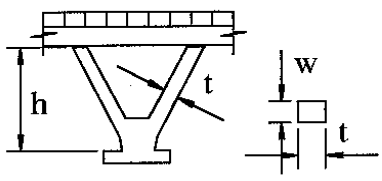
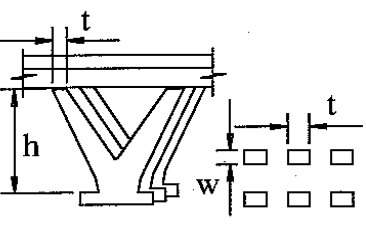
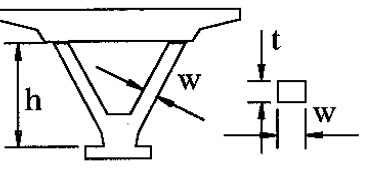
$n = 1$  20 Solid R.C. wall	$n = 1$  21 Cellular R.C. wall	$n = 1$  22 Single R.C. solid column
$n = 1$  23 Single R.C. hollow column	$n = b \cdot v \cdot 3$  24 Multiple R.C. solid round columns	$n = b \cdot v \cdot 4$  25 Multiple R.C. solid columns with beam on top
$w = t = n = n \cdot v \cdot t$  26 Masonary wall	$n = 1$  27 Mass concrete wall	$n = 1$  28 Steel lattice column
$n = 1$  29 Single R.C. splayed pier	$n = b \cdot v \cdot 3$  30 Multiple R.C. splayed piers	$n = 1$  31 Single R.C. V-shape

Figure 4 (a) : Type Piers

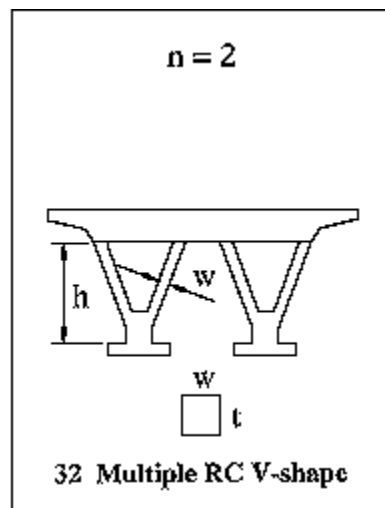


Figure 4(b) : Type Piers

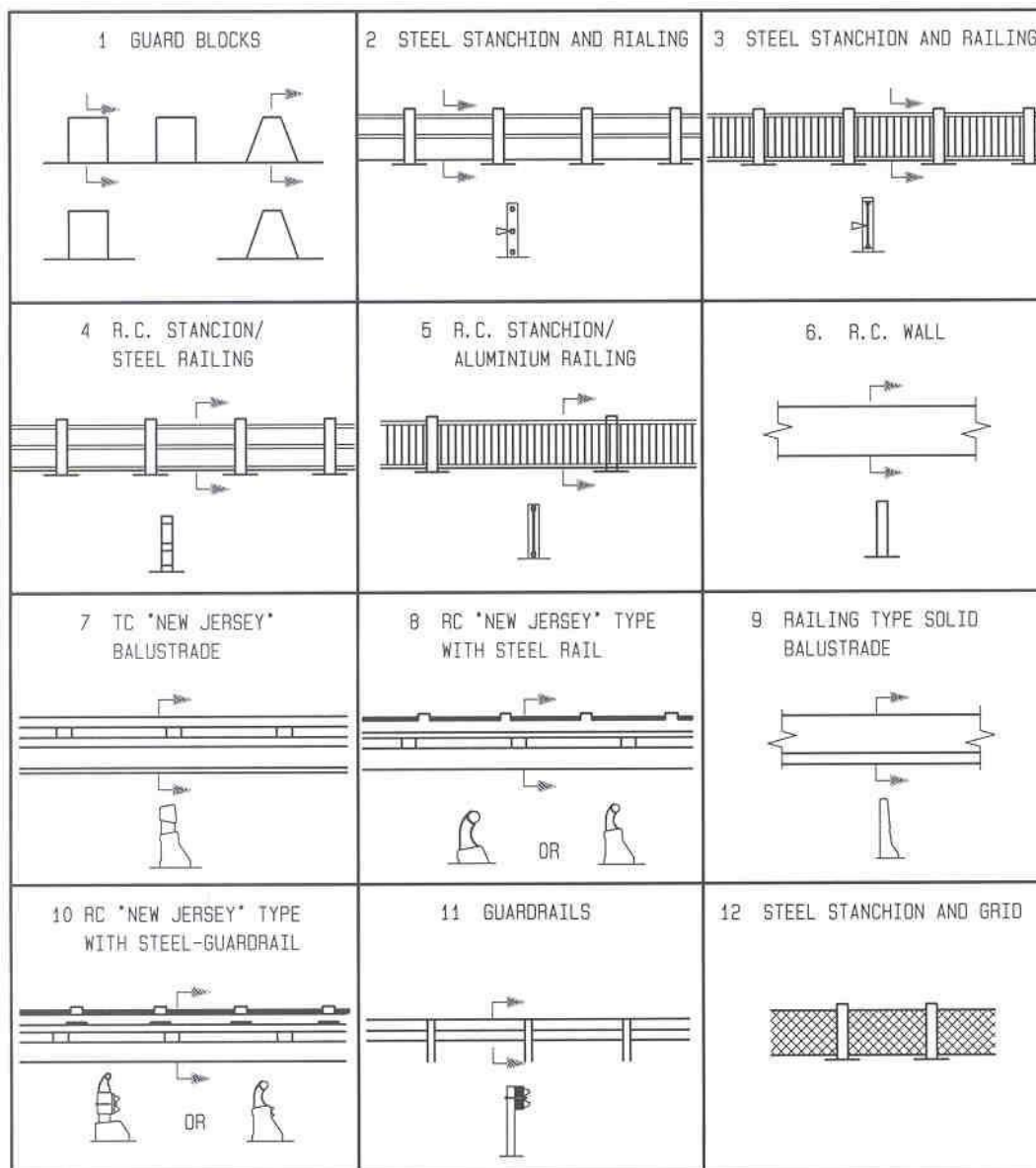


Figure 5: Type parapet/railing

1.22 TYPE WEARING SURFACE

DESCRIPTION: This field indicates the type of wearing surface on the deck.

FORMAT : Integer String : The Type of wearing surface is described by a two digit code.

01 Premixed asphalt	03 Concrete	98 Other
02 Chip-and-spray	04 Gravel	? Unknown

1.23 TYPE EXPANSION JOINTS

DESCRIPTION: This field indicates the type of expansion joint on a deck.

FORMAT : Integer String : This type of joint is described by a two digit code.

Use the expansion joint type on either side of the main span's deck when there is more than one expansion joint type.

01 Concrete nosing with compression seal	12 Open joint with no steel edge or special nosing
02 Concrete nosing with steel edge and compression seal	13 Bolt down joint
03 Concrete nosing with gland in metal/steel runner (BSP, Glaciers, Maurer etc.)	14 Metal finger joint
04 Concrete nosing with joint filler	15 Sliding steel plates
05 Concrete nosing with silicone sealant	16 Modular joint
06 Elastomeric concrete nosing with compression seal	17 Custom built
07 Epoxy nosing with compression seal	18 Buried under surfacing
08 Epoxy nosing with steel edge and compression seal	19 Steel plate buried under surfacing
09 Open joint with steel edge	20 Asphaltic plug-type joint (Thorma)
10 Open joint with concrete nosing and steel edge	21 Hot-poured bitumen rubber sealant
11 Open joint with concrete nosing only	97 None
	98 Other
	? Unknown

1.24 APPROACH SLAB

DESCRIPTION: This field indicates if the bridge has approach slabs.

Y = Yes

N = No

U = Unknown

1.25 ABUTMENT GALLERY

DESCRIPTION: This field indicates if the bridge has abutment galleries

Y = Yes

N = No

U = Unknown

1.26 TYPE BEARINGS

DESCRIPTION: This field indicates the type of bearings on which the bridge deck is carried.

FORMAT : Integer String : The type of bearing is described by a two digit code.

Use the bearing type which carries the main span when there is more than one bearing type.

01 Pot-bearings/PTFE-Type/Teflon

02 Elastomeric rubber pads

03 Roofing felt / Malthoid (slip membrane)

04 Concrete rocker type

05 Steel roller

06 Steel hinge

07 Concrete hinge

08 Steel/ copper plates

97 None

98 Other

? Unknown

1.27 NUMBER OF SPANS

DESCRIPTION: This field indicates the number of spans.

FORMAT : Integer Field : Range between 1 and 99

This is a compulsory field.

1.28 SPAN

DESCRIPTION: This field indicates the number of each of the spans.

FORMAT : Integer Field : Range between 1 and 99

This is a compulsory field.

Spans are numbered from abutment (L) to abutment (R), for example, span 1 is the first span from the left abutment.

IMPORTANT: For a definition of abutment (L)/(R), refer to "Type Abutment (Left)/(Right)".

1.29 SPAN LENGTH (m)

DESCRIPTION: This field indicates the length of a bridge span.

FORMAT : Decimal String : Range between 0,0 and 999,99

For bridge structures the span length is measured on the centre line of the principal feature from centre line of support to centre line of support. For cell structures the span length is measured perpendicular to the centre line of the culvert and should only be the clear opening width.

1.30 DECK DEPTH AT MIDSPAN (m)

DESCRIPTION: This field indicates the maximum depth of the span at midspan. (See Figure 6)

FORMAT : Decimal String : Range between 0,0 and 9,99

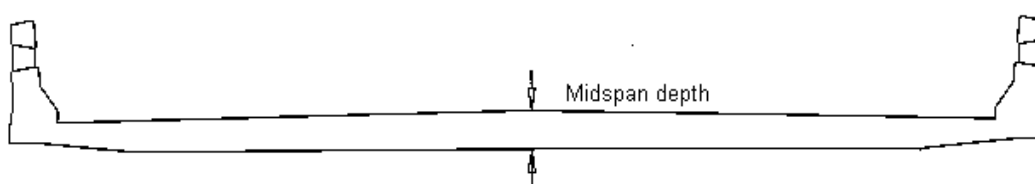


Figure 6 : Midspan depth

1.31 TYPE SPAN

DESCRIPTION: This field indicates the type of span. (See Figure 7 on Page 19).

FORMAT : Integer String : The type of span is described by a two digit code.

01 Simply supported span

02 Continuous span

03 Double cantilevered span

04 Single cantilevered span

05 Single suspended simply supported span	19 Fixed arch - deck at intermediate level
06 Suspended span	20 Lattice girder truss
07 Portal frame	21 Vierendeel truss
08 Cantilevered construction	22 Girdled frame
09 Strut frame	23 Cable stayed bridge
10 Supported cantilevered construction	24 Box culvert (in-situ concrete)
11 Suspension bridge	25 Box culvert (precast concrete)
12 Three hinged arch rib	26 Concrete pipe culvert
13 Three hinged arch slab	27 Corrugated metal pipe culvert
14 Two hinged arch rib	28 Corrugated metal arch (Armco) culvert
15 Two hinged arch slab	29 Concrete arch culvert
16 Bowstring	30 Endspan of continuous type bridge
17 Fixed arch slab	98 Other
18 Fixed arch - deck at crown	? Unknown

1.32 NUMBER OF PIERS

DESCRIPTION: This field indicates the number of piers of a bridge.

FORMAT : Integer String : Range from 0 to 99

This is a compulsory field. A pier can consist of a single column/wall or a row of columns that divides the deck it supports into two spans. Each inner wall of a cell structure (in situ) must be considered a pier. The contacting inner walls of cell structures (precast units) must be considered as one pier.

IMPORTANT : The number of piers is always one less than the number of spans. A single span bridge has 0 (nil) piers

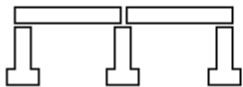
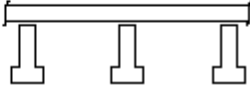
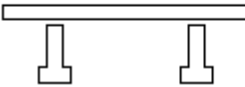
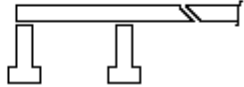
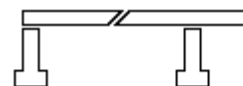
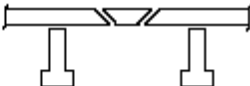
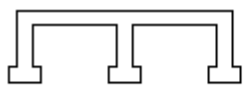
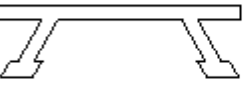
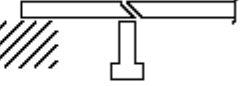
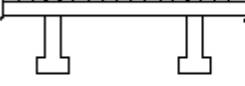
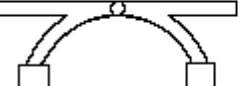
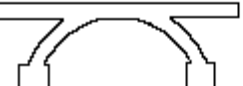
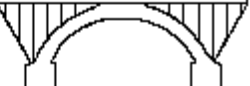
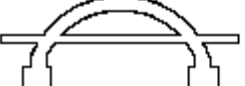
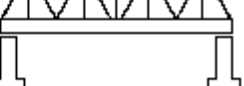
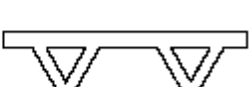
1 Simply supported span 	2 Continuous span 	3 Double cantilevered span 	4 Single cantilevered span 
5 Single suspended simply supported span 	6 Suspended span 	7 Portal frame 	8 Contilevered construction 
9 Strut frame 	10 Supported cantilevered construction 	11 Suspension bridge 	12 Threee hinged arch rib 
13 Three hinged arch slab 	14 Two hinged arch rib 	15 Two hinged arch slab 	16 Bowstring 
17 Fixed arch slab 	18 Fixed arch deck at crown 	19 Fixed arch deck at intermediate level 	20 Lattice girder truss 
21 Vierendeel truss 	22 Girdled frame 	23 Cable stayed bridge 	24 to 29 Culverts 

Figure 7: Type span

1.33 PIER

DESCRIPTION: This field indicates the number of each of the respective piers.

FORMAT : Integer String : Range between 0 and 99

This is a compulsory field. For a bridge with abutments the piers are numbered from abutment (L) to abutment (R) i.e., pier 1 is the first pier adjacent to the left abutment. A pier can consist of a single wall/column or a row of columns that divide the deck it supports in two spans.

1.34 TYPE PIER

DESCRIPTION: This field gives the type of each pier of a bridge. (See Figures 4 (a) on Page 13, and 4 (b) on Page 14).

FORMAT : Integer String : The type pier is described by a two digit code.

This is a compulsory field.

20 Solid R C wall	28 Steel lattice
21 Cellular R C Wall	29 Single R C splayed pier
22 Single R C solid column	30 Multiple R C splayed piers
23 Single R C Hollow column	31 Single R C V-shape
24 Multiple R C solid columns	32 Multiple R C V-shape
25 Multiple R C solid columns (Beam on top)	97 None
26 Masonry wall	98 Other
27 Mass concrete wall	? Unknown

Note: RC = Reinforced Concrete

1.35 NUMBER OF COLUMNS

DESCRIPTION: This field indicates the number of columns at a pier.

FORMAT : Integer String : Range between 0 and 99

1.36 PIER HEIGHT (m)

DESCRIPTION: This field indicates the average height of each pier measured on the centre line of the principal feature. Also see figure 4.

FORMAT : Decimal String : Range between 0,0 and 99,9

The height of a pier column is measured from the top of the footing to the top of the pier (bearings excluded).

1.37 PIER THICKNESS (m)

DESCRIPTION: This field indicates the most prominent thickness of each pier, measured perpendicular to the width of the pier. Also see figure 4.

FORMAT : Decimal String : Range between 0,0 and 99,9

For piers with tapered columns, the dimensions are taken at the top end.

1.38 PIER WIDTH (m)

DESCRIPTION: This field indicates the width of each pier. Also see figure 4.

FORMAT : Decimal String : Range between 0,0 and 99,9

Width refers to the dimension that is generally parallel to the secondary feature. For culverts, the pier width is measured on the centre line of the culvert between the outer edges of the headwalls.

1.39 TYPE FOUNDATION (FOUND)

DESCRIPTION: This field indicates the type of foundation of each pier.

FORMAT : Integer String : The type of foundation is described by a two digit code.

01 Spread footing	05 Cell structure (in situ)
02 Reinforced concrete piles	06 Cell structure (precast slab)
03 Caissons	98 Other
04 Anchored footing	? Unknown

1.40 TYPE FOUNDING MATERIAL FOR PIERS (FOUND MAT)

DESCRIPTION: This field indicates the type of material that the foundation is founded upon.

FORMAT : Integer String : The type of founding material is described by a two digit code.

01 Hard sound rock (10 MPa)	03 Very soft rock (1 MPa)
02 Soft rock (3 MPa)	04 Cemented sand and boulders (700 kPa)

- | | |
|---|--|
| 05 Cemented gravel and coarse sand (600 kPa) | 12 Soft shale (300 kPa) |
| 06 Cemented coarse and fine sand (500 kPa) | 13 Very stiff & dry consolidated clay (300 kPa) |
| 07 Compact gravel and coarse sand (500 kPa) | 14 Stiff clay (150 kPa) |
| 08 Compact coarse sand (400 kPa) | 98 Other |
| 09 Compact fine sand (300 kPa) | ? Unkown |
| 10 Hard shale (700 kPa) | |
| 11 Medium hard shale (500 kPa) | |

1.41 TYPE BEARINGS ON PIERS (ROW 1)/(ROW 2)

DESCRIPTION: This field indicates the type of bearings on which the bridge deck is carried at each pier.

FORMAT : Integer String : The type of bearing is described by a two digit code.

If the bridge deck is continuous over a pier, there will only be one-row bearings per pier (side elevation). Bearings (row 2) will be coded **97**. If there is a joint over a pier, there will generally be two rows of bearings per pier (side elevation). Bearings (in row 1) is on the left abutment side and bearing (in row 2) on the right abutment side of the pier.

- | | |
|---|--------------------------------|
| 01 Pot-bearings/PTFE-Type/Teflon | 07 Concrete hinge |
| 02 Elastomeric rubber pads | 08 Steel/ copper plates |
| 03 Roofing felt / Malthoid (Slip membrane) | 97 None |
| 04 Concrete rocker type | 98 Other |
| 05 Steel roller | ? Unknown |
| 06 Steel hinge | |

1.42 TYPE ABUTMENT LEFT/RIGHT

DESCRIPTION: This field indicates the type of abutment. (See Figure 3 on Page 12)

FORMAT : Integer String : The type of abutment is described by a two digit code.

THE FOLLOWING DEFINITION IS VERY IMPORTANT!

Definition for Abutment (left) and Abutment (right)

When looking along the principal feature in the direction of increasing km distance, abutment (Left) will be at the lower km distance and abutment (Right) will be at the higher km distance. If only the secondary feature has km distance markings, then abutment (Left) is to the left when viewing in the direction of increasing km distance and abutment (Right) to the right.

- | | |
|--------------------------------------|---|
| 01 Mass concrete gravity type | 05 Seating beam/stub column |
| 02 Reinforced concrete | 06 Frame (only bridges, not cell structures) |
| 03 R C wall with RC wingwalls | 07 Buttressed concrete wall |
| 04 Spill through | 08 Masonry wall |

- | | |
|-----------------------------|--|
| 09 Integral pile cap | 22 Solid RC cantilever/wingwall |
| 10 Reinforced earth | 23 Solid RC cantilever/returnwall |
| 11 Counterfort | 97 None |
| 12 Cell structure | 98 Other |
| 13 Multiple V-type | ? Unknown |
| 14 Perched abutment | |

1.43 ABUTMENT HEIGHT (m)

DESCRIPTION: This field indicates the average height of each abutment measured on the centre line of the principal feature. Also see figure 3.

FORMAT : Decimal String : Range between 0,0 and 99,9

The height of an abutment is measured as shown in figure 3 (bearings excluded).

1.44 TYPE ABUTMENT FOUNDATION L/R

DESCRIPTION: This field indicates the type of abutment foundation. The abutment foundation (L) to the left abutment and the abutment foundation (R) to the right abutment.

FORMAT : Integer String : The Type Abutment Foundation is described by a two digit code.

- | | |
|-------------------------------------|---|
| 01 Spread footing | 05 Cell structure (in situ slab) |
| 02 Reinforced Concrete piles | 06 Cell structure (precast slab) |
| 03 Caissons | 98 Other |
| 04 Anchored footing | ? Unknown |

1.45 TYPE FOUNDING MATERIAL FOR ABUTMENTS (FOUND MAT)

DESCRIPTION: This field indicates the type of material that the foundation is founded upon.

FORMAT : Integer String : The type of founding material is described by a two digit code.

01 Hard sound rock (10 MPa)	10 Hard shale (700 kPa)
02 Soft rock (3 MPa)	11 Medium hard shale (500 kPa)
03 Very soft rock (1 MPa)	12 Soft shale (300 kPa)
04 Cemented sand and boulders (700 kPa)	13 Very stiff & dry consolidated clay (300 kPa)
05 Cemented gravel and coarse sand (600 kPa)	14 Stiff clay (150 kPa)
06 Cemented coarse and fine sand (500 kPa)	98 Other
07 Compact gravel and coarse sand (500 kPa)	? Unkown
08 Compact coarse sand (400 kPa)	
09 Compact fine sand (300 kPa)	

1.46 TYPE BEARINGS ON ABUTMENTS

DESCRIPTION: This field indicates the type of bearings on which the bridge deck is carried at each abutment.

FORMAT : Integer String : The type of bearing is described by a two digit code.

01 Pot-bearings/PTFE-Type/Teflon	07 Concrete hinge
02 Elastomeric rubber pads	08 Steel/ copper plates
03 Roofing felt/Malthoid (Slip membrane)	97 None
04 Concrete rocker type	98 Other
05 Steel roller	? Unknown
06 Steel Hinge	

1.47 NUMBER OF EXPANSION JOINTS

DESCRIPTION: This field indicates the number of expansion joints.

FORMAT : Integer Field : Range between 0 and 99

This is a compulsory field.

1.48 JOINT # AND POSITION

DESCRIPTION: These fields indicate the number and position of each of the expansion points.

FORMAT : Joint # : Integer Field : Range between 0 and 99

Position: Text Field	AL	= At left abutment
	AR	= At right abutment
	P1	= At pier number 1
	S1	= In span number 1

Expansion joints are numbered from abutment (L) to abutment (R).

IMPORTANT: For a definition of abutment (L)/(R), refer to "Type Abutment (Left)/(Right)".

1.49 TYPE EXPANSION JOINT

DESCRIPTION: This field indicates the type of expansion joint.

FORMAT : Integer String : This type of joint is described by a two digit code.

01 Concrete nosing with compression seal	12 Open joint with no steel edge or special nosing
02 Concrete nosing with steel edge and compression seal	13 Bolt down joint
03 Concrete nosing with gland in metal/steel runner (BSP, Glaciers, Maurer etc.)	14 Metal finger joint
04 Concrete nosing with joint filler	15 Sliding steel plates
05 Concrete nosing with silicone sealant	16 Modular joint
06 Elastomeric concrete nosing with compression seal	17 Custom built
07 Epoxy nosing with compression seal	18 Buried under surfacing
08 Epoxy nosing with steel edge and compression seal	19 Steel plate buried under surfacing
09 Open joint with steel edge	20 Asphaltic plug-type joint (Thorma)
10 Open joint with concrete nosing and steel edge	21 Hot-poured bitumen rubber sealant
11 Open joint with concrete nosing only	97 None
	98 Other
	? Unknown

1.50 JOINT LENGTH (m)

DESCRIPTION: This field indicates the length of an expansion joint.

FORMAT : Decimal String : Range between 0,0 and 999,99

The expansion joint length is measured on the centre line of the joint.

1.51 COMPLETION DATE

DESCRIPTION: Use this field to indicate the year when construction was completed. This date can be obtained from the bridge number (#) plates during inspection or from as-built drawings.

Note: Use the "As Built" date on the drawings when the bridge number (#) plates are not available. The completion date is to be given in the format: YYYY.

1.52 DATE MAJOR REHABILITATION

DESCRIPTION: This field indicates the completion date of major rehabilitation work performed on the bridge.

FORMAT : Integer String : YYYY

The date of major rehabilitation works is to be indicated in the format YYYY. Rehabilitation is defined as work aimed at upgrading the general condition of a bridge or the load carrying capacity of a bridge (e.g. widening). Please give a description of the rehabilitation in the block for additional remarks.

1.53 WEIGHT/LOAD RESTRICTION

DESCRIPTION: This field indicates if a load restriction is imposed at the bridge or not.

FORMAT : Character String : Enter Y, N or U.

Enter one of the following in the designated space:

Y – Yes

N – No

U - Unknown

If yes, then provide the restriction in ton.

FORMAT : Integer String : Range between 0 and 99999.

$$1\text{m}^3/\text{s} = 1\text{ft}^3/\text{s} \times 0.3048^3$$

1.54 SCOUR PROTECTION

DESCRIPTION: This field indicates the type of scour protection.

FORMAT : Integer String : Two digit code

01 Gabion boxes

02 Gabion mattresses

03 Gabion boxes and mattresses

04 Stone pitching

05 Dolos or tripods

06 Riprap

07 Precast pile curtain

10 None

98 Other

? Unknown

1.55 ANGLE BETWEEN PIER AND RIVER FLOW (DEGREES)

DESCRIPTION: This field indicates the angle between pier and river flow in degrees. It should be close to zero.

FORMAT : Integer String : Range between 0 and 99

1.56 RISK OF SCOUR

DESCRIPTION: This field indicates if there is a risk of scour.

Enter one of the following in the designated space:

Y – Yes

N – No

U - Unknown

1.57 DESIGN SCOUR DEPTH (m)

DESCRIPTION: This field indicates the design scour depth

FORMAT : Decimal String

1.58 MAX SCOUR DEPTH (m)

DESCRIPTION: This field indicates the maximum scour depth.

FORMAT : Decimal String

1.59 OVERALL BRIDGE LENGTH (m)

DESCRIPTION: This field indicates the overall length of the bridge. (See Figure 8).

FORMAT : Decimal String : Range between 0,0 and 999,9.

The bridge length should be measured between outside joint centrelines, along the road centre line. For pipe/cell culverts the bridge length is measured between the outsides of the first and last pipe/cell.

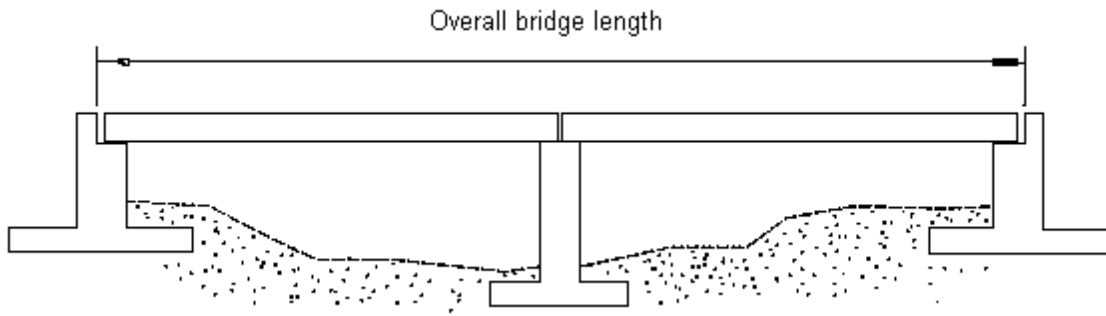


Figure 8: Overall bridge length

1.60 OVERALL BRIDGE WIDTH (m)

DESCRIPTION: This field indicates the overall width of a structure. (See Figure 9)

FORMAT : Decimal String : Range between 0,0 and 999,9.

Overall width is measured from outer edge to inner edge of the deck cross section. The width of the cell structures is measured perpendicular to the centre line of the road between the outer edges of the culvert headwalls.

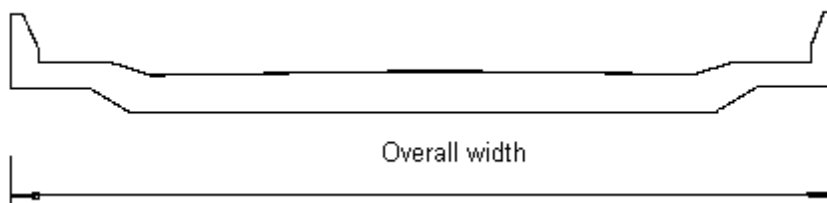


Figure 9: Overall bridge width

1.61 WIDTH BETWEEN PARAPETS/RAILINGS (m)

DESCRIPTION: This field indicates the overall width of a structure between parapets or railings.

FORMAT : Decimal String : Range between 0,0 and 999,9.

This dimension is measured from inner edge to inner edge of the parapets/railings. The width of the cell structures is measured perpendicular to the centre line of the road between the inner edges of the guard-rails or between the shoulder break points.

1.62 MINIMUM BRIDGE ROADWAY WIDTH (m)

DESCRIPTION: This field indicates the tar width of the roadway on the bridge. (See Figure 10)

FORMAT : Decimal String : Range between 0,0 and 999,9.

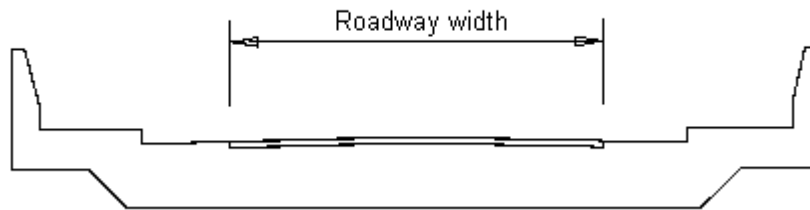


Figure 10: Minimum bridge roadway width

1.63 APPROACH ROADWAY WIDTH (m)

DESCRIPTION: This field indicates the tar width of the approach road to the bridge (obtained from the typical section of the approach roadway). (See Figure 11).

FORMAT : Decimal String : Range between 0,0 and 999,9.

The roadway width must be measured from outer shoulder to outer shoulder for unsurfaced roads.

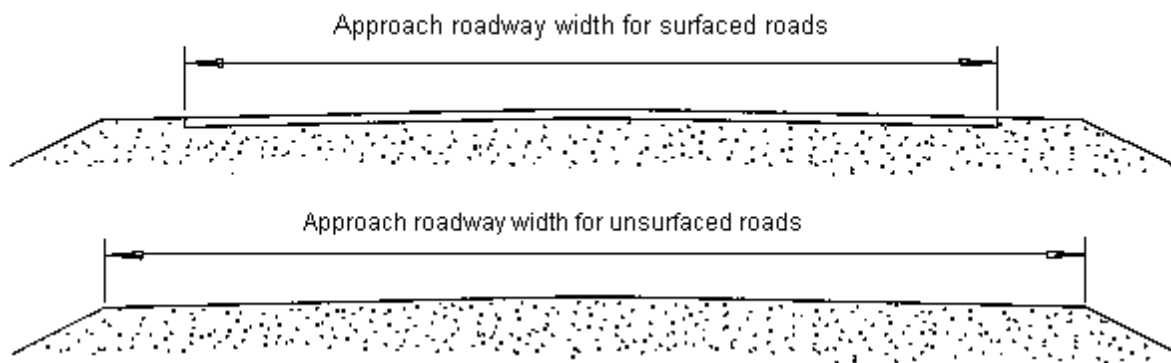


Figure 11: Approach roadway width

1.64 ANGLE OF SKEW

DESCRIPTION: The angle of intersection should be entered in degrees. (See Figure 12)

FORMAT : Integer String : DD.

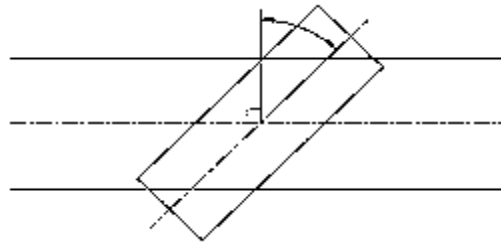


Figure 12: Skew Angle

1.65 RADIUS (m)

DESCRIPTION: This field indicates the horizontal radius length of a bridge on a curve.

FORMAT : Integer String : Range between 0 and 99999.

Bridges on a straight line will be indicated as 99999 or a blank.

1.66 VERTICAL CLEARANCE PRINCIPAL (m)

DESCRIPTION: This field indicates the vertical clearance measurement on the principal feature.

FORMAT : Decimal String : Range between 0,00 and 99,99.

Note: If there is no restriction, please indicate by entering 99.

If the feature passing over the bridge is not a road, this field must be left blank.

1.67 HORIZONTAL CLEARANCE PRINCIPAL (m)

DESCRIPTION: This field indicates the horizontal clearance measurement on the bridge. (See Figure 13).

FORMAT : Decimal String : Range between 0,00 and 99,99.

The horizontal clearance is the limiting measurement for a vehicle to pass over the bridge. The shape of kerbs, balustrades and traffic barriers will determine where the measurement is to be taken. For cell structures the horizontal clearance is the formation width, or the width between guard-rails. If the feature passing over the bridge is not a road, this field must be left blank.

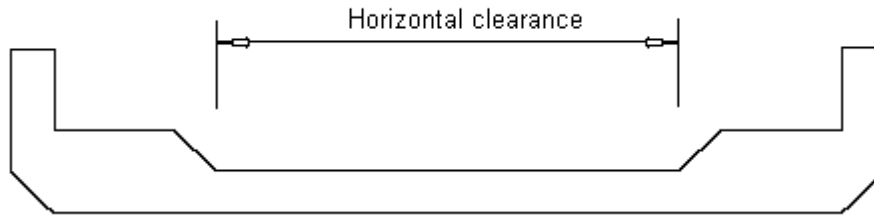


Figure 13: Horizontal clearance

1.68 VERTICAL CLEARANCE SECONDARY (m)

DESCRIPTION: This field indicates the vertical clearance of the road passing under the bridge. (usually at the tie point).

FORMAT : Decimal String : Range between 0,00 and 99,99.

When not obtained from the drawings, the vertical clearance must be measured at the critical point. That is where the distance between the road surface, and the deck soffit is the smallest. Vertical clearance for the secondary feature does not apply for river bridges and drainage structures. If the feature passing under the bridge is not a road, this field must be left blank.

1.69 HORIZONTAL CLEARANCE SECONDARY (m)

DESCRIPTION: This field indicates the horizontal clearance measurement under the bridge. (See Figure 14).

FORMAT : Decimal String : Range between 0,00 and 99,99.

The horizontal clearance must be measured as the distance between the supports/embankment perpendicular to the centre line of the road under the bridge. Horizontal clearance for the secondary feature does not apply for river bridges and drainage structures. If the feature passing under the bridge is not a road, this field must be left blank.

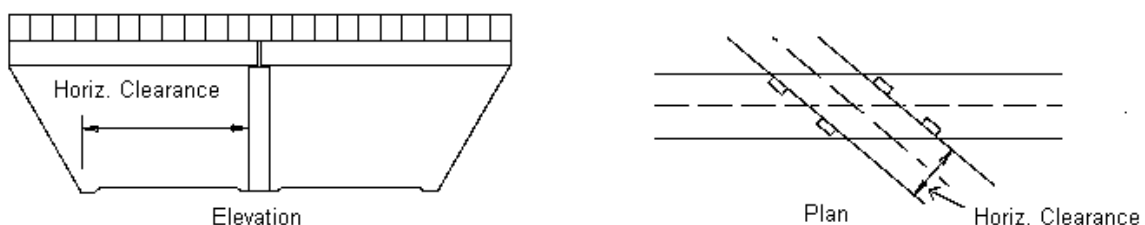


Figure 14: Horizontal clearance – secondary feature

1.70 NUMBER OF LANES

DESCRIPTION: This field indicates the number of traffic lanes over a bridge.

FORMAT : Integer String : Range between 0 and 9.

A fully developed turning lane on a bridge is also considered as a lane.

1.71 TRAFFIC FLOW DIRECTIONS

DESCRIPTION: This field indicates whether a bridge carries one-way or two-way traffic.

FORMAT : Integer String : Either 1 or 2.

1.72 ROAD CLASS (PRINCIPAL/SECONDARY)

DESCRIPTION: This field gives the class of road (Geometric standard classification as used in the RMS) of the road carried by the bridge or the road passing under the bridge.

FORMAT : Character Field : Maximum length of 3 characters.

1.73 NUMBER OF SIDEWALKS

DESCRIPTION: This field indicates the number of sidewalks over a bridge.

FORMAT : Integer String : Range between 0 and 9.

1.74 AVERAGE DAILY TRAFFIC (PRINCIPAL/SECONDARY)

DESCRIPTION: This field indicates the average daily traffic (vehicles per day) passing over (principal) or under (secondary) the bridge.

FORMAT : Integer String : Range between 0 and 9999999.

Average daily traffic is measured in equivalent vehicle units per day. A heavy* vehicle is counted as equivalent to three standard vehicles.

*Axle load > 4000 kg, or gross vehicle mass > 7000 kg, or carrying capacity > 3000 kg.

1.75 TRUCK FACTOR (%) (PRINCIPAL/SECONDARY)

DESCRIPTION: This field indicates the percentage of the traffic volume that is made up of heavy vehicles.

FORMAT : Integer String : Range between 0 and 99%

The following values can be used as a guide-line when information is unavailable:

- High percentage (%) of heavy vehicles: 20%.
- Average percentage (%) of heavy vehicles: 8%.
- Low percentage (%) of heavy vehicles: 2%.

1.76 YEAR OF COUNT PRINCIPAL/SECONDARY

DESCRIPTION: This field indicates the year in which the traffic count was done.

FORMAT : Integer String : YYYY

The year in which the traffic count was done is to be given in the format: YYYY (for example 1993). When an estimated value is used for the ADT, use the year that the estimate was done.

1.77 ACCESS – PIERS

DESCRIPTION: This field indicates the access to all the piers for field inspections.

FORMAT : Integer String : Two digit code

- 01** No piers
- 02** Easy access - binoculars adequate
- 03** Located in rivers - require boat
- 04** Require bush cleaning
- 05** Require inspection vehicle
- 06** Inaccessible
- 98** Other
- ?** Unknown

1.78 ACCESS – BEARINGS

DESCRIPTION: This field indicates the access to all the bearings for field inspections

FORMAT : Integer String : Two digit code

- 01** No bearings
- 02** Easy Access - binoculars adequate
- 03** Require 6 m ladder
- 04** Access via manhole in deck box girder soffit
- 05** Inaccessible (bricked up)
- 06** Inaccessible (covered by steel plates)
- 07** Inaccessible (encased in polystyrene)

- 08 Inaccessible (buried below ground)
- 09 Inspection vehicle required
- 98 Other
- ? Unknown

1.79 ACCESS – DECK SOFFIT

DESCRIPTION: This field indicates the access to all the deck soffits for field inspections

FORMAT : Integer String : Two digit code

- 01 Easy Access - binoculars adequate
- 02 Require inspection vehicle
- 03 Require bush clearing
- 04 Boat required
- 05 Inaccessible
- 98 Other
- ? Unknown

1.80 ACCESS – ABUTMENT GALLERIES

DESCRIPTION: This field indicates the access to the abutment galleries for field inspections.

FORMAT : Integer String : Two digit code

- 01 No galleries
- 02 Manhole in road shoulder
- 03 Gallery doors locked with padlock (require key)
- 04 Gallery doors require special spanner to open
- 05 Gallery doors require angle grinder to open
- 06 Inaccessible (welded closed, bricked up)
- 98 Other
- ? Unknown

1.81 ACCESS – INSIDE OF BOX GIRDER

DESCRIPTION: This field indicates the access to the inside of a box girder for field inspections.

FORMAT : Integer String : Two digit code

- 01 No box girders
- 02 Inaccessible voids or solid deck section

- 03** Abutment gallery doors
- 04** Hole in deck box girder soffit - require unbolting
- 05** Abutment manhole in road shoulder
- 98** Other
- ?** Unknown

SECTION 2

2.1 OVERALL RATE CALCULATION

The Overall Rate Calculation formula is used to calculate an overall rating for a bridge from the five condition ratings for the five bridge components (approach, waterway, substructure, superstructure, and roadway). The Overall Rate is calculated once the principal bridge inspection condition ratings have been entered and the “Calculate Overall Rate” button is pressed (BMS: Capture Screens / Add inspection / Final).

To calculate the Overall Rate it is important that:

- If the specific rating for any of the five components is not known because some of the components do not exist, then an average rating for these components must still be estimated. Tick the NA blocks for the specific components. Do not use a default rating of 9 (excellent) where a component does not exist.
- If the specific rating for any of the five components is not known because some of the components could not be rated due to being inaccessible, then an average rating for these components must still be estimated. Add a comment in this regard in the comments column. Do not use a default rating of 0 (beyond repair) where a component is not accessible and could not be seen to be rated.

Use the Data Validation Reports from the BMS to identify bridges with outstanding main component ratings (BMS / Data validation Reports / Outstanding Main Component Rating).

The weighted factors per rating for each of the five bridge component condition ratings are shown below.

(BMS: System Parameters / Overall Rating Variables)

Category	0	1	2	3	4	5	6	7	8	9
Approaches	6	6	6	2	2	0.25	0.25	0.25	0.25	0.25
Waterway	8	8	8	2	2	1	1	1	1	1
Substructure	8	8	8	4	4	2	2	2	2	2
Superstructure	8	8	8	4	4	2	2	2	2	2
Roadway	6	6	6	2	2	0.5	0.5	0.5	0.5	0.5

(0 = Beyond repair, 9 = Excellent)

Table 2.1: Weighted factors per rating for Overall Rate calculations

The Overall Rate is rounded to an integer before being used to categorize bridges into 5 condition categories, i.e. critical, poor, marginal, satisfactory, good, as follows:

Calculated Overall Rate	Condition Category
0 and 1	Critical
2 and 3	Poor
4 and 5	Marginal
6	Satisfactory
7 to 9	Good

Table 2.2: Categorization of Overall Rates into condition categories

2.2 THE RANKING FORMULA

The Overall Deficiency ranking formula is used to rank bridges for specific purposes, based on user definable criteria that include the horizontal clearance, vertical clearance, approach alignment and bridge condition. The user can also indicate if traffic volumes and strategic importance should be considered in the ranking formula.

The BMS User Manual describes the ranking as follows:

“A primary objective of the system is to identify those bridges that need attention. In order to do this, bridges are ranked according to a deficiency index. The index is made up of three basic components, as follows:-

- ***The Level of Service Provided to the Public***

The degree to which a bridge meets user demands is based on an evaluation of the following physical properties of the bridge:

- Vertical Clearance.
- Horizontal Clearance.
- Roadway alignment over and/or under the bridge.

- ***Condition of the Bridge***

Maintaining a bridge in an acceptable condition incurs costs. The lack of maintenance may also lead to user costs, if the condition of the bridge deteriorates to the point where the load capacity of the bridge is affected, or where joints may damage vehicles.

- ***Importance of the Bridge***

The relevant importance of each bridge can be measured in terms of the traffic volumes that cross the bridge and the strategic importance of the bridge. Strategic importance is measured in terms of the potential delays in traveling time if the bridge were to be closed.

The contribution of each of the above aspects in the calculation of the deficiency index is governed by a weighting factor assigned to each aspect. The different weighting factors can be changed to suit the particular emphasis of an investigation.”

The Overall Condition Deficiency Index is given by the following expression:

$$D = D_{HC} + D_{VC} + D_{AA} + D_C$$

D = 100 indicates a bridge with major defects, D = 0 indicates a bridge with only minor or no defects.

Where:

D_{HC} = Horizontal clearance deficiency

D_{VC} = Vertical clearance deficiency

D_{AA} = Approach alignment deficiency

D_C = Bridge condition deficiency;

And:

$$D_{HC} = W_{HC} \cdot F_1 \cdot \left[\frac{ADT_O}{ADT_B} \right]^{K_1} \cdot \left[\frac{W_G - W_O}{W_G - W_M} \right]^{K_2}$$

$$D_{VC} = W_{VC} \cdot \left[\frac{ADT_U}{ADT_B} \right]^{K_1} \cdot \left[\frac{VC_G - VC_O}{VC_G - VC_m} \right]^{K_2}$$

$$D_{AA} = W_{AA} \cdot \left[\frac{ADT_O}{ADT_B} \right]^{K_1} \cdot K_{AA}$$

$$D_C = W_C \cdot \left[\frac{ADT_O}{ADT_B} \right]^{K_4} \cdot F_{SI} \cdot \sum_{i=1}^n k_i \cdot w_i$$

Where:

W_{HC} = Weighting factor for horizontal clearance deficiency

W_{VC} = Weighting factor for vertical clearance deficiency

W_{AA} = Weighting factor for approach alignment deficiency

W_C = Weighting factor for bridge condition deficiency

(A weighting can thus be placed on the 4 deficiencies, depending on whether all or only some or only one is to be included in the ranking. The four weightings should add up to 100. If only the bridge condition has to be considered in the ranking, then $W_c = 100$ and the others are all 0)

And:

For horizontal clearance deficiency:

W_{HC} = Weighting factor for horizontal clearance deficiency (see above)

F_1 = Number of lanes factor

ADT_O = Average daily traffic over the bridge

ADT_B = Average daily traffic base volume

W_G = Desirable width

W_M = Minimum acceptable width

W_O = Actual width

K_1 = A constant

K_2 = A constant

And:

For vertical clearance deficiency:

W_{VC} = Weighting factor for vertical clearance deficiency (see above)

ADT_U = Average daily traffic under the bridge

ADT_B = Average daily traffic base volume

VC_G = Desirable height (clearance)

VC_M = Minimum acceptable height (clearance)

VC_O = Actual height (clearance)

K_1 = A constant

K_2 = A constant

And:

For approach alignment deficiency:

W_{AA} = Weighting factor for approach alignment deficiency

ADT_O = Average daily traffic over the bridge

ADT_B = Average daily traffic base volume

K_1 = A constant

K_{AA} = Approach alignment condition rating coefficient

And:

For bridge condition deficiency:

W_C = Weighting factor for bridge condition deficiency

ADT_O = Average daily traffic over the bridge

ADT_B = Average daily traffic base volume

K_4 = A constant

F_{SI} = Strategic importance factor

k_i = Bridge component (i) condition rating coefficient

w_i = Weighting factor for bridge component (i) condition rating

n = number of components

(Condition rating coefficients for all bridge components are considered here, ie superstructure, substructure, roadway, approach and waterway, based on their weights.)

To calculate the Overall Condition Deficiency Index it is important to note that:

- If the data as required for a specific deficiency (D_{HC} or D_{VC} or D_{AA} or D_C) to be included in the ranking is not known, then it should not be included.
- The data requirements for the five condition ratings for the five bridge components have to be met (refer to “Overall Rate Calculation” regarding the data requirements for the approach, waterway, substructure, superstructure and roadway).

Use the Data Validation Reports from the BMS to identify bridges with outstanding data (BMS / Data validation Reports / Outstanding Main Component Rating and Insufficient Data for Ranking).

The values or expressions for the above variables are set out in the following tables and formulas:

2.2.1 NUMBER OF LANES FACTOR : F_1

(BMS: System Parameters / Ranking Variables / Number of Lanes Factor)

Total number of lanes	Two way traffic (single)	One way traffic (dual)
1	1,0	0,8
2	1,0	0,8
3	1,0	0,6
4	0,8	0,4
5	0,8	0,4
6	0,6	0,4
7	0,6	0,4
8	0,6	0,4
9	0,6	0,4
10	0,6	0,4

Table 2.3: The number of lanes factor F_1 as a function of the number of lanes and directions of traffic (used in Horizontal Clearance Deficiency)

2.2.2 MINIMUM ACCEPTABLE CLEAR DECK WIDTH (W_M)

(BMS: System Parameters / Ranking Variables / Horizontal Clearance Limits)

Road Class	Goal lane width (m)	Minimum lane width (m)	Description
1.0	3.7	3.7	Throughway - 2 carriage ways
2.0	3.7	3.7	Conventional 2 carriage way road
3.0	3.7	3.7	Single carriage way throughway
3.1	3.7	3.7	High standard single carriage way paved road
3.2	3.7	3.4	Average standard single carriage way paved road
3.3	3.4	3.0	Low standard single carriage way paved road
4.1	3.4	3.4	Gravel road (Fair alignment)
4.2	3.4	3.4	Gravel road (Less than Fair alignment)
4.3	0	4	Gravel road (Less than Fair alignment)
4.4	0	3	Gravel road (Less than Fair alignment)
4.5	0	3	Gravel road
Def	3.7	3.4	Default

Table 2.4: Goal lane width (W_G) and the minimum acceptable width (W_M) as a function of the class of road (used in Horizontal Clearance Deficiency)

(Note the specific road class codes being used here, these codes should also be used in the database for road classification.)

2.2.3 VERTICAL CLEARANCE LIMITS

(BMS: System Parameters / Ranking Variables / Vertical Clearance Limits)

Road Class	Desirable clearance (m)	Minimum clearance (m)	Description
1.0	5.2	4.7	Throughway - 2 carriage ways
2.0	5.2	4.7	Conventional 2 carriage way road
3.0	5.2	4.7	Single carriage way throughway
3.1	5.1	4.7	High standard single carriage way paved road
3.2	5.1	4.7	Average standard single carriage way paved road
3.3	5.1	4.7	Low standard single carriage way paved road
4.1	5.1	4.7	Gravel road (Fair alignment)
4.2	5.1	4.7	Gravel road (Less than Fair alignment)
4.3	5.1	4.7	Gravel road (Less than Fair alignment)
4.4	5.1	4.7	Gravel road (Less than Fair alignment)
4.5	5.1	4.7	Gravel road
Def	5.2	4.7	Default

Table 2.5: Vertical clearance limits as a function of the class of road (used in Vertical Clearance Deficiency)

(Note the specific road class codes being used here, these codes should also be used in the database for road classification.)

2.2.4 LIST OF CONSTANTS

(BMS: System Parameters / Ranking Variables / General Constants)

ADT _B	=	5400
K ₁	=	0,4
K ₂	=	1,5
K ₃	=	1,5
K ₄	=	0,2

Table 2.6: List of constants used in the ranking formula

2.2.5 WEIGHTING FACTORS FOR BRIDGE COMPONENTS (W_i)

(BMS: System Parameters / Ranking Variables / Weighted Factors per Bridge Components)

Category	W _i
Superstructure	1,00
Substructure	1,00
Roadway	0,50
Approach	0,25
Waterway	0,83

Table 2.7: Weighting factors (W_i) for each bridge component in the Bridge Condition Deficiency Formula (used in bridge condition deficiency)

2.2.6 BRIDGE COMPONENT CONDITION RATING COEFFICIENT (K_i)

(BMS: System Parameters / Ranking Variables / Bridge Condition Rating Coefficient)

Condition Rating	Super Structure	Sub Structure	Roadway	Approaches	Waterway
0	1,0	1,0	1,0	1,0	1,0
1	1,000	1,000	1,000	1,000	1,000
2	1,000	1,000	1,000	1,000	1,000
3	0,500	0,500	0,625	0,700	0,425
4	0,250	0,250	0,425	0,450	0,200
5	0,100	0,100	0,250	0,250	0,100
6	0,025	0,025	0,125	0,100	0,030
7	0,000	0,000	0,000	0,000	0,000
8	0,000	0,000	0,000	0,000	0,000
9	0,000	0,000	0,000	0,000	0,000

Table 2.8: Bridge component condition rating coefficients (K_i) for each component and condition rating (used in bridge condition deficiency)

2.2.7 APPROACH ALIGNMENT COEFFICIENT (K_{AA})

(BMS: System Parameters / Ranking Variables / Approach Alignment Deficiency)

Condition Rating	Coefficient K _{AA}
0	1.0
1	1,0
2	1,0
3	1,0
4	0,5
5	0,2
6	0,0
7	0,0
8	0,0
9	0,0

Table 2.9: Approach alignment coefficients (K_{AA}) for different condition ratings (used in approach alignment deficiency)

2.2.8 STRATEGIC IMPORTANCE FACTOR (F_{si})

Expression for F_{SI}

$$F_{SI} = f_c \cdot \left[\frac{t}{60} \right]^{K_3} + 1.0$$

Where

t = detour impact on travel time (minutes)

K₃ = constant

f_c = factor for substructure/superstructure condition rating.

(BMS: System Parameters / Ranking Variables / Sub-Superstructure Condition Rating)

Substructure/Superstructure condition rating	Coefficient f_c
0	1.09
1	1,09
2	1,09
3	1,09
4	0,43
5	0,17
6	0,04
7	0,04
8	0,04
9	0,04

Table 2.10: Coefficient (FC) for the sub- and superstructure condition rating (used in bridge condition deficiency)

The following limits and conditions are set:

$$0 \leq D_{HC} \leq W_{HC}$$

$$0 \leq D_{VC} \leq W_{VC}$$

$$0 \leq D_{AA} \leq W_{AA}$$

$$0 \leq D_C \leq W_C$$

$$0,3 \leq \left[\frac{ADT_{O/U}}{ADT_B} \right]^{K_1} \leq 2,5$$

$$0,7 \leq \left[\frac{ADT_O}{ADT_B} \right]^{K_4} \leq 1,5$$

Summary:

Deficiency to be included in calculation of Overall Deficiency ranking	Requirements for data in BMS	Selections
Horizontal clearance deficiency	- ADT for all bridges (ADT and truck factor) (see * below at Traffic) - Actual horizontal clearances for all bridges	$W_{HC} > 0$ (in System Parameters / Ranking Variables / Ranking Criteria)
Vertical clearance deficiency	ADT under bridge for relevant over-road bridges (ADT and truck factor) (see * below at Traffic)	$W_{VC} > 0$ (in System Parameters / Ranking Variables / Ranking Criteria)

	• Actual vertical clearances for all relevant bridges	
Approach alignment deficiency	• ADT for all bridges (ADT and truck factor) (see * below at Traffic) • All approach alignments to be rated	$W_{AA} > 0$ (in System Parameters / Ranking Variables / Ranking Criteria)
Bridge condition deficiency	• ADT for all bridges (ADT and truck factor) (see * below at Traffic) • All bridge component condition ratings, even if not accessible to rate or not existent	$W_C > 0$ (in System Parameters / Ranking Variables / Ranking Criteria)
Strategic importance factor	• Not operational	Tick (in System Parameters / Ranking Variables / Ranking Criteria) (not operational)
Traffic	• ADT for all bridges and truck factor	Tick (in System Parameters / Ranking Variables / Ranking Criteria) (* if not ticked, then traffic is also not required for the other deficiency calculations)